Lock N Key: Exploring Rotation-Based Authentication in Virtual Reality

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1 Introduction

Virtual Reality (VR) has grown significantly in recent years, and more users are starting to use VR for things beyond just gaming. People are using it for social interaction, productivity, and even accessing personal or sensitive information. As VR becomes more common in daily life, one challenge that does not get talked about enough is authentication. How does a user securely log in or enter a password while inside a virtual environment? This is a problem worth exploring because the input methods we rely on in the real world, like keyboards and touchscreens, do not work the same way in VR.

Current VR authentication options like virtual keyboards and PIN pads are not ideal. They tend to be slow, uncomfortable to use, and can be vulnerable to people watching you input your password [7, 21]. Some researchers have looked into alternatives like gaze-based input and 3D password systems [2, 7], but there is still a lot of room to explore what kinds of interactions feel most natural in VR. Since users in VR are already grabbing and rotating objects as part of normal interaction, using rotation as a form of input seemed like a natural direction to take.

For this project, we are building a system called *Lock N Key*, where users enter a password by rotating a 3D cube in a specific sequence using their VR controllers. Each rotation in a certain direction counts as one step in the password. The goal is to see whether this kind of interaction is something users can learn, remember, and feel comfortable using. We will be looking at things like how users naturally try to interact with the cube, how well they can memorize a rotation-based password, and whether this feels like a practical alternative to other VR input methods.

2 Related Work

Authentication in VR has been studied from a few different angles, and most of the work points to the same general conclusion: traditional input methods like PIN pads and virtual keyboards do not translate well into immersive environments. George et al. [9] were among the first to evaluate this directly, testing PIN and pattern-based authentication in VR and finding that while both methods remained usable, they were significantly harder to observe than in the physical world. Building on this, George et al. [8] showed that using 3D space for authentication is more resistant to observation attacks than 2D methods, with attackers only succeeding 12.5 percent of the time in their RoomLock system. Later work by Mathis et al. [21] and Bu et al. [2] built on this by showing that physical manipulation of 3D objects during authentication makes it significantly harder for observers to track what a user is entering. Riyadh et al. [24] also found that authentication methods designed natively for VR, like gesture-based input, consistently outperform methods ported from 2D interfaces like PIN pads in both speed and usability. Yu et al. [28] reached a similar conclusion earlier, finding that while 3D password systems had the highest security against shoulder surfing, PIN and pattern systems were perceived as more usable, highlighting the ongoing trade-off between security and usability in VR authentication.

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The choice of input method also plays a big role in how well these systems work. George et al. [7] found that combining gaze for pointing with a controller trigger for selection was the fastest and most secure combination they tested. Mathis et al. [21] similarly found that controller tapping was the fastest input method while gaze offered the strongest observation resistance. Olade et al. [23] explored porting the popular SWIPE pattern system into VR and found that the hand-held controller was the most usable input method and that VR environments were highly resistant to shoulder surfing compared to mobile devices. Khamis et al. [13] further showed that mid-air gestures are more secure than touch but slower, while gaze offers the strongest resistance to video-based attacks. Funk et al. [6] proposed LookUnlock, which uses spatial and virtual targets on HMDs as passwords and found that virtual passwords were completely resistant to shoulder surfing in their evaluation. Across these studies, a broader review by Noah and Das [22] confirmed that no single method excels in all areas and that there is still a need for approaches that better balance security and usability in VR.

What none of these studies explore is using the physical rotations of a 3D object as the password itself. Most systems use a 3D object as a tool to select digits or other targets [19, 21], or use the 3D environment to place objects that users point at [7, 8, 27]. Some work has explored gesture-based passwords in mixed reality environments such as picture passwords on the HoloLens [11], and Gurary et al. [10] demonstrated that 3D authentication on the Oculus Rift can exceed the password space of traditional alphanumeric passwords with better memorability. More recently, Zheng et al. [29] proposed VRFlipAuth, which uses cube-flipping operations as a password in VR and found it to be fast and highly resistant to shoulder surfing. Abdelrahman et al. [1] also showed that controller trackpad input significantly outperforms motion controller input for VR authentication. Our system takes a different approach where the sequence of rotations a user performs on a cube is the password, which is a form of interaction that feels natural in VR and has not been directly studied before. Eiband et al. [4] documented that shoulder surfing is a real and common threat in everyday life, which further motivates our goal of designing an authentication method that is both native to VR and inherently resistant to observation.

We imagined a system different from the typical typing or entry of digits in a normal password system as shown above, which can be a struggle for many users in VR, and wanted to attempt a system which pulled upon a more typical VR input method using grabbing and gesturing for password entry.

Similar articles and work has examined VR passwords and authentication through different types of review and other biometrics. We have found several different articles have focused on biometrics, but leave a gap in usability [3, 12, 25, 26]. Reseachers have explored many different options using VR headsets, with some articles stating that eye movement patterns may be sufficient to act as a biometric just using gaze [16, 17]. And combining eye movement/ gaze with brainwave data further improves authentication accuracy and security [5]. Other physiological signals unique to the VR have also been explored: OcuLock [18] uses electrooculography to measure low-level human visual system responses, SoundLock [30] uses auditory-evoked pupillary responses, and SigA [14] demonstrates that remote photoplethysmography captured by a face-oriented microcamera inside the headset can authenticate users from cardiac signals. Mathis et al. [20] also used VR itself as a research tool, showing that simulated environments can produce valid usability results on par with traditional lab settings. Behavioral biometrics derived from body motion during ordinary VR tasks have also been shown to identify users with up to 90% accuracy [15]. Together, these approaches show the wide variety of authentication strategies available in VR or extended reality, and how authentication methods that avoid biometric data collection remain a not entirely explored topic.

3 Methodology

3.1 Participants

Participants were recruited to participate in a user study that examined the usability and memorability of a multi-modal spatial password authentication system implemented in virtual reality. All participants were required to provide their informed consent prior to participating in the study.

3.2 Technology Used

The experiment was conducted using a Meta Quest 2 head-mounted display (HMD), a standalone virtual reality headset. The virtual environment was developed using the Unity game engine (version [6.3]), which provided the interactive 3D simulation in which the password task was performed.

3.3 Password System Design

The spatial password system consisted of a 3×3 grid of nine identical cubes arranged in a flat plane within the virtual environment. These were placed world-fixed, within the field of vision of the participant. Each cube could be manipulated independently in two different ways:

- **Horizontal movement:** Each cube could be displaced forward or backward, locking into a set position if pushed forward or backward.
- **Rotational movement:** Each cube could be rotated about its vertical axis in 90-degree increments, clockwise (right) or counterclockwise (left).

The combination of horizontal positions and orientations across all nine cubes determined the password state. A unique password was defined as a specific configuration of cube positions and orientations within the grid.

3.4 Procedure

Each participant completed the study in a single session. Upon putting on the Meta Quest 2 headset and receiving a brief orientation to the virtual environment and the cube manipulation interface, participants were sequentially presented with three distinct passwords to enter. For each password, the target password was communicated to the participant, who was then required to replicate the configuration by interacting with the 3×3 cube grid within the virtual environment.

Three passwords were presented in a fixed order, and users were given a familiarization period prior to the formal task to acclimate to the interaction mechanics. Each password increased in difficulty compared to the last. No time constraints were imposed during password entry, however completion time was recorded.

3.5 Data Collection

Following the completion of all three password entry tasks, participants' feedback and reflective responses were recorded. Qualitative data were gathered through a semi-structured interview and a post-experiment questionnaire, in which participants were asked to reflect on their experience using the password system. The topics addressed included perceived ease of use, memorability of the password configurations, sense of presence within the virtual environment, and overall satisfaction with using the system.

3.6 Analysis

PLACEHOLDER FOR NOW - DON'T KNOW IF THIS NEEDS TO GO HERE YET.

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